

Article

Classification of Oil Palm Fresh Fruit Bunches Based on Their Maturity Using Thermal Imaging Technique

Shahrzad Zolfagharnassab ¹, Abdul Rashid Bin Mohamed Shariff ^{1,2,3,*}, Reza Ehsani ⁴, Hawa Ze Jaafar ⁵ and Ishak Bin Aris ⁶

¹ Department of Biological and Agricultural Engineering, Engineering Faculty, Universiti Putra Malaysia, Serdang 43400, Selangor, Malaysia

² Smart Farming Technology Research Center (SFTRC), Universiti Putra Malaysia, Serdang 43300, Selangor, Malaysia

³ Institution of Plantation Studies (IKP), Universiti Putra Malaysia, Serdang 43300, Selangor, Malaysia

⁴ Department of Mechanical Engineering, School of Engineering, University of California, Merced (UC Merced), 5200 North Lake Rd., Merced, CA 95343, USA

⁵ Department of Crop Science, Agriculture Faculty, Universiti Putra Malaysia, Serdang 43400, Selangor, Malaysia

⁶ Department of Electrical and Electronic Engineering, Engineering Faculty, Universiti Putra Malaysia, Serdang 43400, Selangor, Malaysia

* Correspondence: rashidpls@upm.edu.my

Citation: Zolfagharnassab, S.; Shariff, A.R.B.M.; Ehsani, R.; Jaafar, H.Z.; Aris, I.B. Classification of Oil Palm Fresh Fruit Bunches Based on Their Maturity Using Thermal Imaging Technique. *Agriculture* **2022**, *12*, 1779. <https://doi.org/10.3390/agriculture12111779>

Academic Editor: Jiangbo Li

Received: 19 September 2022

Accepted: 21 October 2022

Published: 26 October 2022

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Copyright: © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Abstract: The maturity of oil palm Fresh Fruit Bunches (FFB) is considered to be a significant factor that affects the profitability and salability of palm oil FFB. Typical methods of grading FFB consist of physical grading of fresh fruit, which is time-consuming and expensive, and the results are prone to human error. Therefore, this research attempts to formulate a thermal imaging method to indicate the precise maturity of oil palm fruits. A total of 297 oil palm FFBs were collected. The samples were divided into three groups: under-ripe, ripe, and over-ripe. Afterward, all the samples were scanned using a thermal imaging camera to calculate the real temperature of each sample. In order to normalize the measurement, the difference between the average temperature of the palm bunch and the ambient temperature (ΔTemp) was considered as the main parameter. The results indicated that the mean ΔTemp of oil palm FFBs decreased consistently from under-ripe to over-ripe. The results of the ANOVA test demonstrated that the observed significance value was less than 0.05 in terms of ΔTemp , so there is a statistically significant difference in the means of all three maturity categories. It can be concluded that ΔTemp is a reliable index to classify the FFBs of oil palm. The classification analysis was conducted using the ΔTemp of the FFBs and its application as an index in Linear Discriminant Analysis (LDA), Mahalanobis Discriminant Analysis (MDA), Artificial Neural Network (ANN), and Kernel Nearest Neighbor (KNN). The highest degrees of overall accuracy (99.1% and 92.5%) were obtained through the ANN method. This study concludes that thermal images can be used as an index of oil palm maturity classification.

Keywords: artificial neural network (ANN); principal component analysis (PCA); oil palm fresh fruit bunches; thermal imaging

1. Introduction

The oil palm tree, also known as “*Elaeis guineensis* Jacq”, is a plant that grows well in the tropical region. The oil palm tree was developed and used as an agricultural crop, and currently, it has become the most important commodity crop in Malaysia, being the world’s second largest producer of palm oil, generating up to 18.12 million tons of crude palm oil from a planted area of 5.74 million hectares (more than two-thirds of its agricultural land) in 2021. The country’s total export of palm oil and palm-based products was 24.28 million tons [1].

A serious challenge encountered at the time of processing fruits and extracting oil is deemed to be grading the oil palm FFBs with regard to their maturity for maximum and quality yield extraction. With that in mind, classifying the oil palm FFBs into certain maturity categories seems to be an influential factor that underlies milling operations in an efficient manner. It is of high importance to take the quality of oil palm into consideration since it influences both the refining process and the end product [2].

Currently, the oil palm FFB is harvested manually, with maturity being guided by FFB color and the number of loose fruitlets on the ground. Harvesting is done when FFBs turn into orange-red or once the fruitlets on the ground are identified and counted. A fruitlet is an individual and single fruit of oil palm FFB (Figure 1).

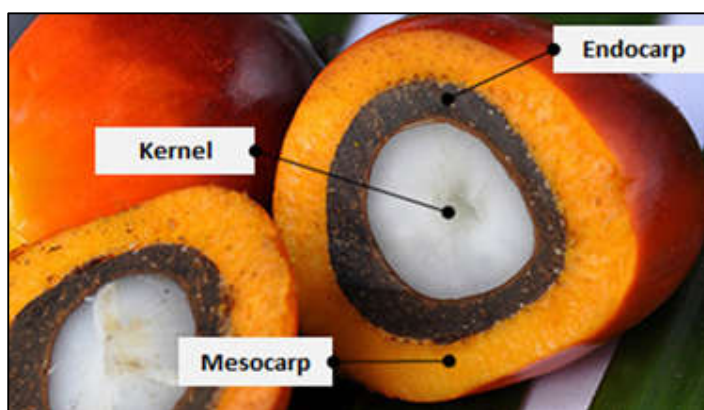


Figure 1. The characteristics of individual oil palm fruitlets (Vijaya et al., 2009) [3].

There have been several research studies on potential methods to determine the maturity of palm fruits. In order to determine and assess the level of maturity of oil palm fruits, some researchers have made use of hyperspectral imaging [4,5], fluorescence sensing [6,7], computer vision and an optical sensor [8,9], inductive sensing [10,11], near-infrared spectroscopy [12], laser imaging [13], and lidar scanning [14].

Advanced color-based machine vision was able to categorize distinct fruitlets into correct groups with an accuracy of 90% [15,16]. In the same line of study, optical properties and an associated system have already been developed. The system uses color intensity, segmentation, and disparate color spaces to specify oil palm FFBs across the spectrum of maturity. The authors of [17] presented a method that classified oil palm FFBs according to their maturity using reflectance data obtained from a four-band sensor. In addition, a pattern recognition algorithm was employed for categorizing the oil palm FFBs into the three respective classes (un-ripe, ripe, and over-ripe). Similarly, [6] made use of the Multiplex® multi-parameter fluorescence sensor. The mentioned device paved the ground for classifying oil palm FFBs according to flavonoids and anthocyanin. The authors of [18] developed the ANN model to classify the ripeness of oil palm fruitlets using the Raman spectroscopy. The result shows 95.48% classification accuracy. There were some limitations in [18] in which the classification was not for oil palm bunches. The research in [19] proposed developing an automated detection model using a deep learning method to classify oil palm FFB based on maturity level. They used several types of YOLO algorithms. The result showed that YOLOv4 has a higher accuracy of 98.70% in classifying the maturity of oil palm FFB.

The authors of [20] used Raman spectroscopy for maturity classification of oil palm fruitlets based on the molecular assignments extracted from the Raman bands of 1240 cm^{-1} and 1360 cm^{-1} . They deployed an automated classification system based on the ANN algorithm with an overall performance of 97.9% accuracy.

Although there are many research studies on oil palm FFB grading, there is still room to use new methods such as thermal imaging to improve grading accuracy and

decrease the complexity of high-dimensional computation. The next section introduces some research studies that used thermal imaging in the agricultural field.

Thermal imaging has been used for different applications in agriculture, especially in pre-harvesting and post-harvesting processes [21]. Assessment of seedling viability, estimation of soil water status, estimation of crop water status, irrigation scheduling, identification of diseases and pathogenic plants, estimation of fruit yield, and ultimately, evaluation of the maturity of fruits and vegetables are a few applications of thermal imaging in agriculture [22]. Many researchers have investigated thermography and its role in evaluating the maturity of fruits. In this regard, [23] utilized infrared thermometry to assess the maturity of persimmon, pear, and tomato. They categorized the collected samples into the following three grades: immature, mature, and over-ripe. To do so, color, firmness, and sugar content were taken into account. Since the difference in surface temperature during maturity stages was small, the samples were stored at high (30 °C) and low (5 °C) temperatures for more than 24 h. The surface temperature of immature fruits kept at a lower temperature appeared to be somehow higher than that of mature and over-ripe fruits, and the surface temperature of immature fruits kept at a higher temperature appeared to be somehow lower than that of mature and over-ripe fruits. They claimed that “the heat generated in metabolic processes of fruits and vegetables influence surfaces temperatures of the samples”. The authors of [21] used a commercial infrared thermal imaging camera (VARIOSCAN 2011) to particularize the maturity of two varieties of apples. Their findings indicate that as the degree of maturity increases, the surface temperature of the samples will increase as well. Interestingly, transpiration turns out to decrease during this interval of the fruit ripening process. For fruit maturity estimation, infrared technology can be used to classify the whole fruit into immature and ripe states without considering the color of the fruit. This comes from the fact that ripe fruit have higher heat capacity, and therefore, their body temperature slowly changes [24]. Evaluation of maturity is considered to be highly crucial in pre-harvest and post-harvest stages. Despite the fact that there are a number of automatic methods in this regard, visual inspection is a necessity, and it is implemented in many parts of the world [25]. The temperature of the fruit is determined according to transpiration, metabolic heat, and water exchange with the atmosphere [21]. It is a fact that transpiration occurs in fruits during ripening and storage. As a result, it would be possible to recognize temperature differences in ripening stages via thermography techniques. The authors of [24] applied two techniques, namely thermal imaging and NIR imaging, to assess the maturity of a particular type of mango. In a similar vein, the authors of [26] utilized a thermal imaging camera (called the AGEMA 880 LWB) to highlight the differences in thermal properties that exist between apples with and without an affected water core. It should be noted that the affected area, in this case, was the tissue. The aforementioned camera is sensitive within the spectral range of 8–13 μm .

The authors of [27] developed a real-time system to check the surface temperatures of the carrot uniformity. It was intended to disinfect the carrots in the post-harvest stage. The authors of [28] employed an integrated system including a hyperspectral camera and a thermal imaging camera to identify existing bruises in apples. They demonstrated that a controlled categorization of images (within a spectrum range of 400–5000 nm of surface imaging) could boost the process of detecting early bruises with distinct depths. The authors of [29] proposed and showed how thermal imaging cameras could be used to classify fruits into different maturity levels. They demonstrated that, based on the difference in temperature capacity of immature, mature, and over mature-fruits, the thermal imaging technology is very promising in estimating the maturity level of mango. The authors of [30] utilized the capability of thermal imaging to determine the quality of paddy based on the following properties, moisture content, immature condition, and foreign material. Their method gave accurate results with 92% for moisture content determination and 90% for maturity stage prediction. To our knowledge, no research has been reported on studying the physical appearance of FFB and its internal properties

using thermal imaging under actual field operation. The authors of [31] evaluated infrared thermal imaging coupled with machine learning techniques to distinguish the different pineapple cultivars. Several types of machine learning approaches were compared by [31] for the classification of pineapple cultivars. The results showed that support vector machines with the highest classification rate of 100% could be applied to discriminate of pineapple cultivars.

The present study elaborates upon the potentialities of infrared images (i.e., thermal images) coupled with machine learning algorithms as indicators to categorize the maturity of oil palm FFBs., a novel thermal imaging technique is presented as a new method to classify oil palm FFBs into correct maturity categories. This paper seeks to investigate the correlation between the mean temperatures of oil palm FFBs determined by the thermal imaging method with the oil palm FFB maturity categories and to classify the maturity of oil palm FFBs using thermal imaging. The main contribution of this research lies in utilizing non-contact, non-destructive, fast, and simple in-field methods with acceptable classification accuracy for grading FFBs of oil palm.

The remainder of this paper is structured as follows: The materials and methods are presented in the second part, followed by the results and discussion. Finally, conclusions are drawn.

2. Materials and Methods

In this study, data collection was performed at United Plantation Research and Development in Teluk Intan (UPRD), Perak, Malaysia.

The utilized method starts with sample preparation and ends with the selection of the best predictor and classification method. The workflow of the study is given in Figure 2. Two experiments were carried out; the first experiment was done with 147 samples of FFBs, and the second experiment, to validate the findings of the first experiment, was done with 150 samples.

2.1. Samples and Equipment Preparation

A total of 297 fresh fruit bunches of oil palm from the most common cultivar of oil palm bunches known as Nigrescens were collected for the two experiments based on the following three maturity groups: under-ripe, ripe, and over-ripe (Figure 3). The total number of samples representing under-ripe, ripe, and over-ripe categories of oil palm FFBs was 47, 54, and 46, respectively, for the first experiment and 49, 54, and 47, respectively, for the second experiment.

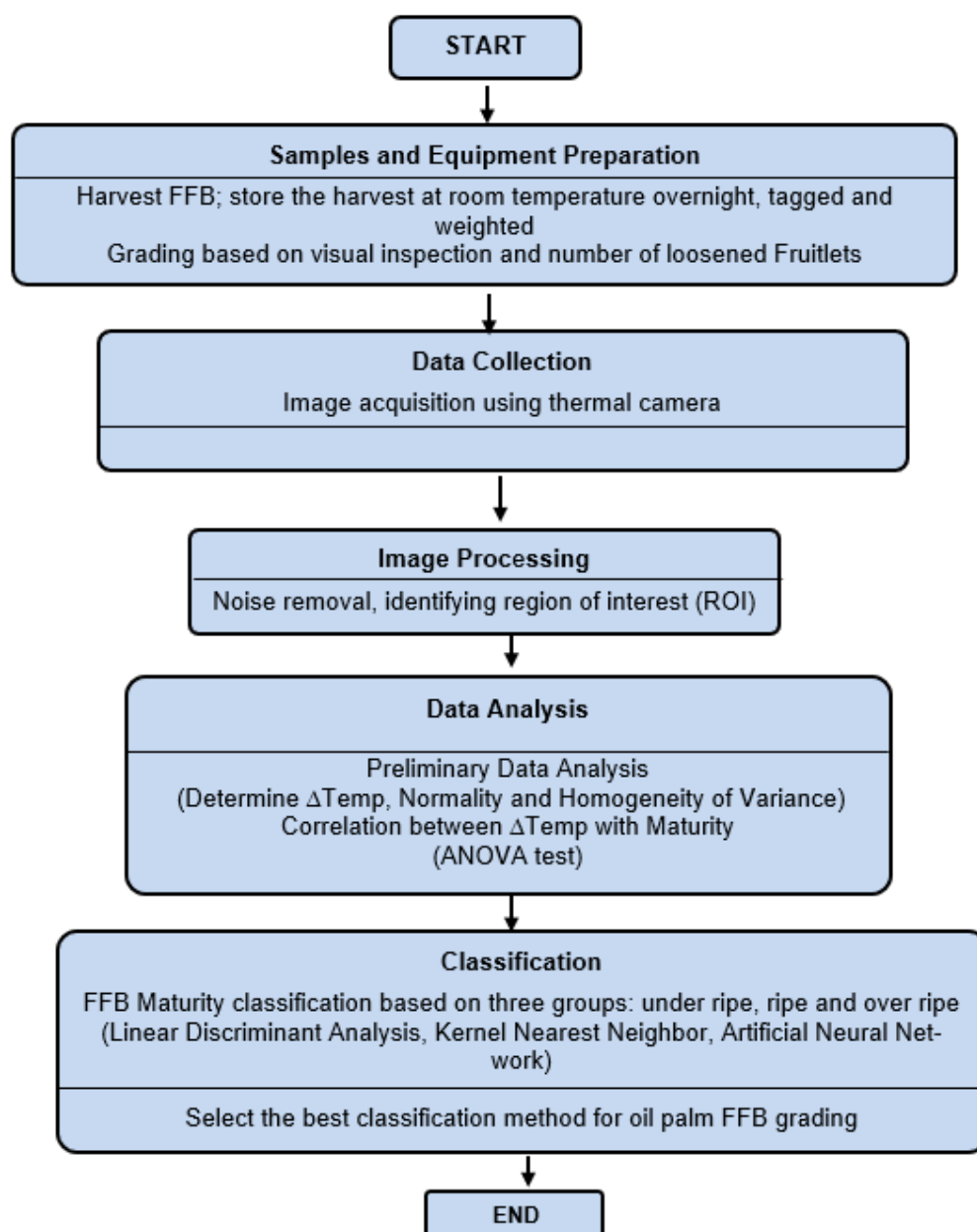


Figure 2. Workflow of research methodology.

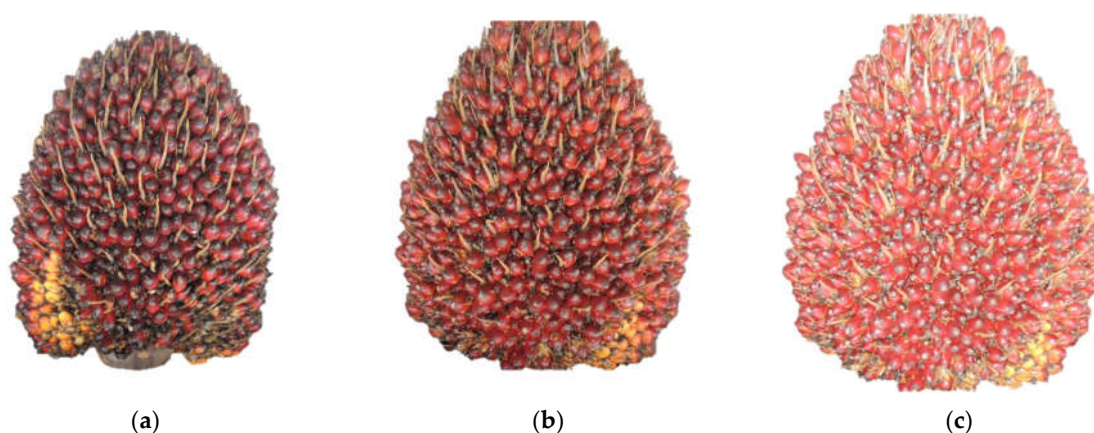


Figure 3. Oil palm fresh fruit bunches of cultivar Nigrescens based on maturity: (a) under-ripe, (b) ripe, and (c) over-ripe.

FFBs samples were harvested from the plantation and kept at room temperature overnight prior to being scanned by the thermal camera. The mentioned samples were tagged, weighted, and classified by trained oil palm graders. Both the standard guideline [32] and the conventional method, which is based on the number of loosened fruitlets, were taken into account (Table 1). Once scanning of the given samples was completed, thermal images of the front and back sides of the FFBs were taken.

Table 1. Oil palm FFB maturity categories are based on the total number of empty sockets representing the total number of loose fruitlets [32].

Total Number of Loose Fruitlets	Maturity Category
0	Under-ripe
0–10	Ripe
>10	Over-ripe

2.2. Thermal Imaging Camera

FLIR E60 thermal camera, (FLIR-USA) was used to record the thermal image data from the oil palm FFB. The FLIR E60 offers 320×240 resolution with an excellent 0.05°C thermal sensitivity. The entire image range covers 76,800 pixels. This thermal camera can measure temperatures ranging from -4 to 1202°F (from -20 to 650°C). In order to measure temperature accurately, the thermal image camera should be calibrated for the following parameters: the emissivity of the object, distance, relative humidity, atmosphere temperature, and surrounding radiation.

The emissivity of the object: Emissivity is regarded as the most influential parameter and must be set correctly. It refers to the amount of radiation that emits from an object in comparison with that from a black body at the same temperature. All objects with a temperature greater than absolute zero emit thermal radiation. The emitted thermal radiation can be measured and considered as an index of the temperature of that particular object. The authors of [33] determined the emissivity of oil palm fresh fruit bunches in the following three stages of maturity: under-ripe, ripe, and over-ripe. They developed the measurement method to obtain the emissivity of oil palm fresh fruit bunches as an important parameter for thermal imaging. The results showed that the emissivity of oil palm in the three important stages of ripeness (under-ripe, ripe, and over-ripe) is 95%. Hence, in all applications of infrared thermal cameras to quantify the oil palm fresh fruit ripeness and oil content, the emissivity can be set to this value.

Distance: As a parameter, distance refers to the space between the object and the front lens of the camera. The distance was set at a fixed value of one meter for all samples in this study.

Relative Humidity: The nature of the camera reveals that transmittance is dependent upon the relative humidity in the atmosphere. In this study, the humidity was recorded every half hour using a digital humidity meter (hygrometer) and set inside the thermal camera.

Atmospheric Temperature: The camera was able to account for the effect of ambient temperature as well. Accordingly, the ambient temperature was recorded every half hour using a thermometer.

Surrounding Radiation: It is demonstrated that surrounding radiation that reflects on an object will not have any impact on the thermal measurement of that object with high emissivity. However, in the case where emissivity is at a low level and the temperature of the object is somehow distinct from that of the reflected one, it is highly crucial to make up for the reflected temperature in the correct way.

3. Data Collection

This research investigates the potential of infrared images (thermal images) as a predictor to classify the oil palm FFB ripeness.

3.1. Image Processing

The images of oil palm FFB collected using the thermal camera were processed to identify the temperature distributions of the selected areas on the FFB. The main steps of this process are outlined in the following sections.

3.1.1. Noise Removal

The authors of [34] showed that an infrared (IR) camera has a noise of about 80 millikelvins. Another point that must be remembered is that the objects (targets) are normally hotter (several Kelvin) than their backgrounds. Overall, the noise is low, especially in indoor spaces such as rooms and laboratories. In outdoor cases, the acquisition of IR images is influenced by environmental factors. As a result, the atmospheric loss can reduce the temperature of the object significantly, particularly at long distances by means of dust, smoke, and fog. Although this may be true, some objects (targets) are close to their backgrounds in terms of temperature.

In the present study, all captured images were processed with a median filter using ThermCAM Researcher Pro Software. The median filter decreases the noise of images. The underlying assumption is that signal pixels correspond to close relationships with their neighboring pixels in a small area, which is called 'filtering window'. The neighboring information paves the ground for the loss of thin lines and textures. Small objects and targets appear to occupy a few pixels in IR images. In addition, they are enclosed by high-gradient pixels. Therefore, in the course of the filtering process, many of them will be substituted by medial pixels and, by extension, they will be removed from the IR image. The noise removal filter and its role in a sample image are demonstrated in Figure 4.

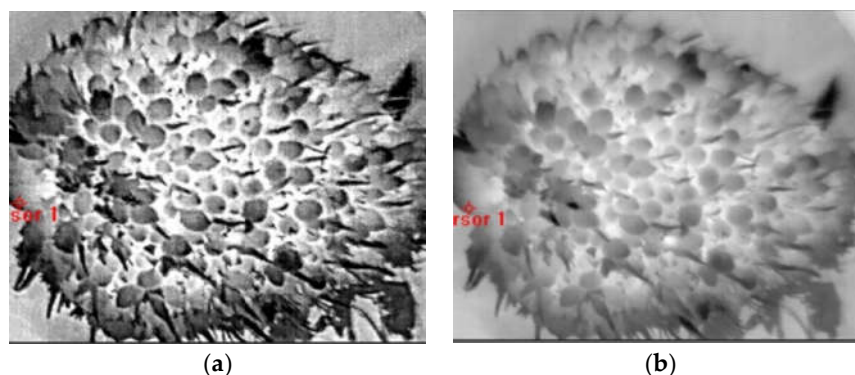


Figure 4. The FFB Image: (a) before and (b) after the noise is removed.

3.1.2. Identifying the Region of Interest (ROI)

The first thing to consider is that the oil palm fruit area needs to be separated from its background. This process begins with recognizing foreground objects (i.e., palm bunches) and background objects. Defining ROI is the first and primary step in thermography processing analysis. ThermCAM Researcher Pro software makes use of regular prismatic shapes, including rectangles, squares, circles, and ellipses, for defining these regions. In this research, the palm bunch was similar to the eclipse in terms of shape. Accordingly, the eclipse was chosen to define the ROI. Both the ROI and the eclipse shape are given in Figure 5.

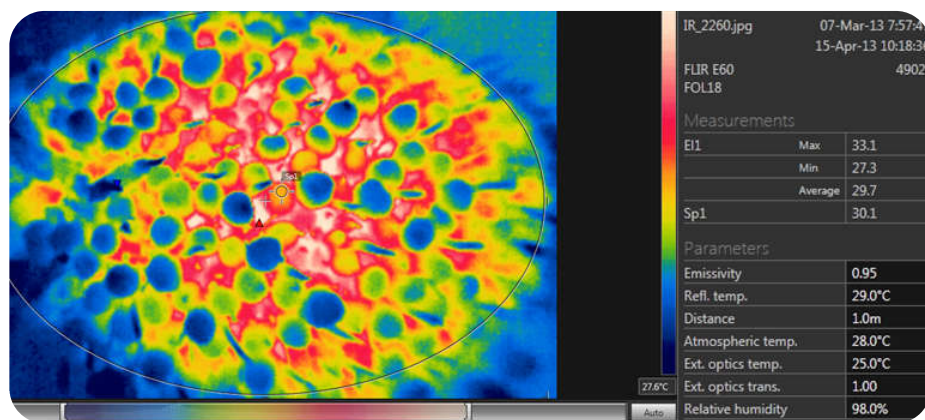


Figure 5. Determination of ROI based on prismatic shapes.

It is the exclusion of irrelevant data and the inclusion of relevant data that matters in evaluating thermal images. Errors and misunderstandings may affect the process of analyzing thermal images. In order to deal with the limitations of irrelevant data in the process of evaluation, image segmentation algorithms were utilized for the intended ROIs and, thereby, they were optimized through the exclusion of irrelevant data.

The ThermCAM Researcher Pro software proposed for thermographic processing allows the user to choose any ROI irrespective of its geometric shape. This software has a segmentation algorithm that is based on the thresholding segmentation method. The given method is the most accurate and precise algorithm for this purpose. The segmentation algorithm tends to optimize the selected region via the removal of those areas that do not have relevant statistical data. As a result, the temperature of the ROI was considered (Figure 6).

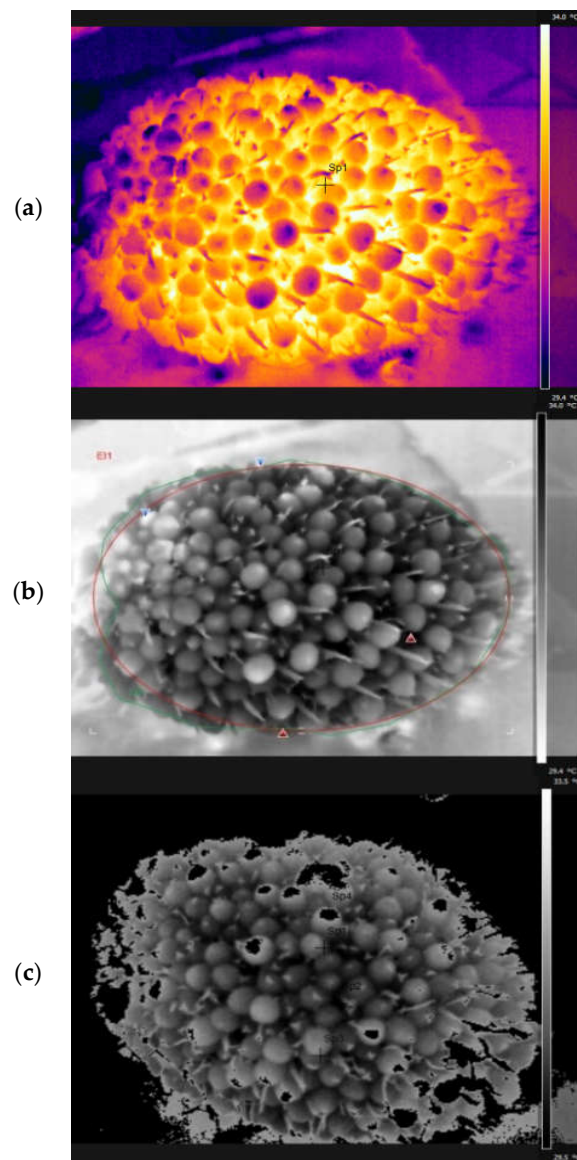


Figure 6. Thresholding segmentation for ROI of an oil palm FFB sample: (a) main thermal image, (b) gray scale image, and (c) segmented image.

Visual investigations have demonstrated that fixed thresholding is able to remove low-intensity regions. Border identification of fruits plays a key role in determining the average temperature accurately. That being the case, boundary digitizing () improves the accuracy of determining the average temperature of the FFB. This method was used for further and more detailed characterization (Figure 7).

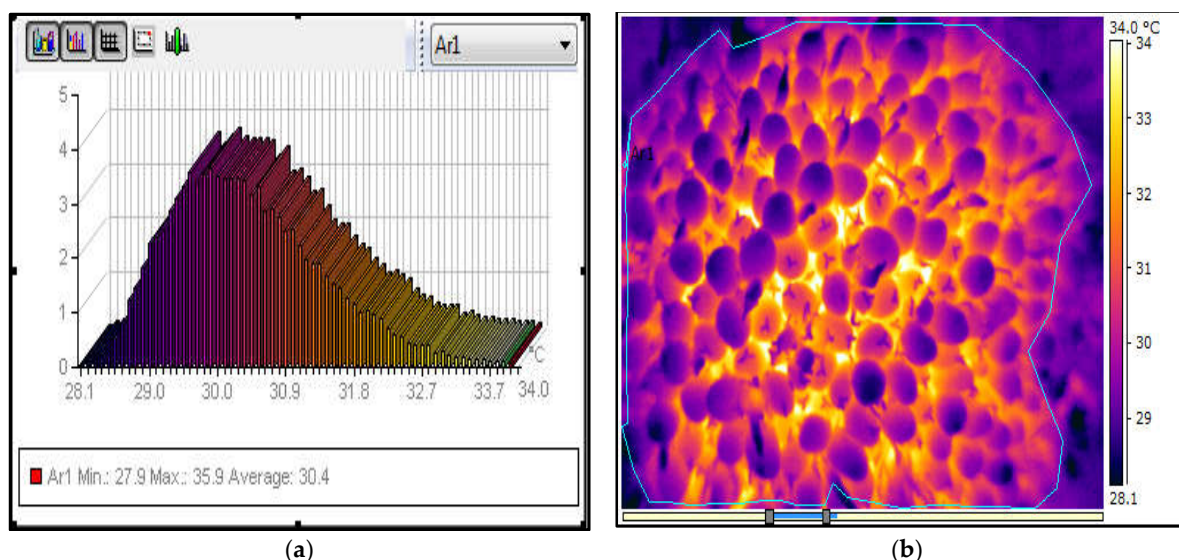


Figure 7. Identifying the ROI: (a) histogram of a thermal image after normalization and (b) ROI on FFB by digitizing method.

3.2. Data Analysis

Data analysis refers to the process of inspecting, cleansing, transforming, and modeling data. It should be noted that such an analysis is done to unearth useful information, reach informative conclusions, and improve decision-making. Data analysis consists of various aspects, approaches, and techniques. It is practical in various disciplines, including business, science, and the humanities.

3.2.1. Preliminary Analysis

The average, maximum, and minimum temperatures of each ROI were calculated. To account for the temperature of the atmosphere, the measurement was calibrated by subtracting the average temperature of the palm bunch from the ambient temperature, which had been recorded throughout the image acquisition process. It is important to realize that ambient temperature changes during the data collection process, and this influences the palm fruit surface temperature.

$$\Delta \text{Temp} = \text{Palm Fruit Temperature} - \text{Ambient Temperature} \quad (1)$$

Therefore, ΔTemp was considered as the main parameter in the present analysis. $\Delta \text{Temperature}$ divided by the weight ($\Delta \text{Temp}/\text{Wt}$) is another parameter in this study.

A box and whisker plot was utilized in the present study to recognize and differentiate the outliers. First and foremost, the outliers were omitted from all datasets, and then they were used for categorization. Levene's test (a test of homogeneity of variance) was conducted to see whether various groups were different or the same. In this case, the null hypothesis is either accepted or rejected. Next, Kolmogorov–Smirnov and Shapiro–Wilk tests were done to ensure that the given data were distributed normally. However, violation of this assumption must be taken into account. More specifically, the Shapiro–Wilk test was conducted in the present study since the dataset appeared to contain less than 2000 components. Moreover, log transformation was implemented for each dataset since it was not normally distributed. Afterward, in order to reinforce the argument that numerical data represent samples from normally distributed populations, a Welch and Brown–Forsythe test was done. The results demonstrated that the variances and sizes of the involved groups were similar, and the groups were considered to be independent.

3.2.2. Correlation between Δ Temp with Maturity

In this study, the variances were analyzed (through the ANOVA test) to see whether the means of dependent variables were different in the involved groups or not. The mean differences and relations between variables can be indicated via the aforementioned analysis. In the same line of thought, PCA was done to assess the correlation between variables, which, in its own turn, helped discover uncorrelated (orthogonal) variables.

3.3. Classification Analysis

This study drew from three classification methods to achieve the highest degree of accuracy. Accordingly, once the features were chosen as predictors, the classification was done as correctly as it is possible. The SPSS software played a key role in the process of classification. The classifiers which were measured in the present study are given as follows: Discriminant Analysis, Kernel Nearest Neighbor, and Artificial Neural Network.

3.3.1. Kernel Nearest Neighbor (KNN)

The KNN is utilized in the present study as a supervised machine learning classifier. KNN relies on the fact that samples in a dataset turn out to exist in close affinity with other samples that have identical features. The mechanism thereof works according to a distance metric of proper similarity.

The nearest neighbor of 'K' is regarded as one of the most engaging algorithms for pattern classification. In practical terms, the selection of 'K' depends on experimental experience. In spite of that, the principle of the nearest neighbor of 'K' draws from a fixed number of nearest neighbors via the feature space and distance method. In the present study, various 'K' values and distance measurement methods were used to balance the classification of FFB maturity. Values and methods with a low confidence level were deleted in line with that balance. To develop the KNN classifier for FFB maturity of oil palm in the grading system, a number of measures were taken to formulate the nearest neighbors of 'K', which are given below:

1. Determining the proper value of 'K', or the number of nearest neighbors;
2. Checking the proper distance metric to measure the distance between the query instance and other training samples;
3. Sorting the distance and specifying the nearest neighbors by with the minimum distance of 'K';
4. Categorizing the nearest neighbors;
5. Using a simple majority of the nearest neighbors as the prediction value for the query instance.

3.3.2. Artificial Neural Network (ANN)

The types of neural networks used in this study are Multilayer Perceptron (MLPs) that are trained with the Back-Propagation algorithm. The typical MLP consists of a number of processing elements (called neurons) that are usually arranged into layers: an input layer, an output layer, and one or more hidden layers. Each processing element in the specific layer is joined to the processing elements of other layers via weighted connections. The input from each processing element in the previous layer is multiplied by an adjustable connection weight. This combined input then passes through a nonlinear transfer function (sigmoid or purelin function) to produce the output of the processing element. The output of one processing element provides the input to the next processing element. In this work, the ANN model is developed with flexible and useful software for this type of application; the IBM SPSS Statistic 20.

The network was composed of some input neurons. A hidden layer includes various neurons and output neurons to illustrate maturity categories. Each hidden neuron computes a function of the weighted sum of its inputs using a sigmoid function. The

sigmoid transfer function was commonly used in the back-propagation networks. It takes the input, which can have any value between plus and minus infinity and squashes the output into the range of 0–1.

One set of networks is used for optimal network structure investigation. A network structure with one input and three outputs. ΔTemp as inputs and three maturity categories (under-ripe, ripe, and over-ripe) as outputs. After the selection of the optimal structure, the ANN was trained with various numbers of samples, and the trained network was used to predict the maturity category.

4. Results and Discussion

The results obtained are presented and discussed as follows: correlation between ΔTemp with FFB maturity; comparing mean of temperature variable based on maturity category; ΔTemp behavior in the FFB maturity process; mean comparison of the first and second experiment; classification analysis of FFB maturity; and classification results comparison.

4.1. Correlation between ΔTemp with FFB Maturity

Under all consideration, data were prepared to guarantee that they are clean and can be utilized for analysis. The value of ΔTemp is given in Figure 8. In addition, the data related to thermal sensors and the box and whisker plots are presented in the aforementioned figure. As it is shown, the outlier samples, represented by the red “*” mark in the box plot of Figure 8a, were deleted from the dataset, after which the dataset was used for further analysis and categorization. In short, an outlier refers to a value that lies outside the inter-quartile range. The present study concentrates on clean data, which, in its own turn, is manifested in the form of a normal distribution (Figure 8b). As a result, the total numbers of samples that lie in the clean data and represent the under-ripe, ripe, and over-ripe categories of oil palm FFBs were calculated for the first and second experiments.

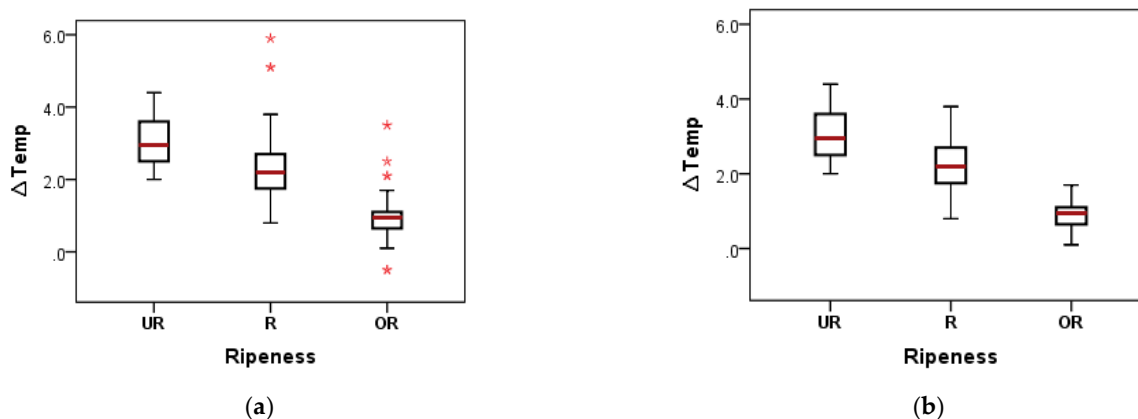


Figure 8. Box and whisker plots represent the ΔTemp of under-ripe, ripe, and over-ripe oil palm FFBs. (a) samples with outliers, (b) clean data, *: outlier samples.

Table 2 shows the total samples used for the analysis. After removing the outliers, there were 141 samples for the first experiment and 149 samples for the second experiment. The numbers of samples for each category (under-ripe, ripe, and over-ripe) and both experiments are shown in Table 2.

Table 2. Total number of samples before and after removal of outliers.

Category	First Experiment		Second Experiment	
	Raw Data	Clean Data	Raw Data	Clean Data
Under-ripe	47	44	49	49
Ripe	54	53	54	53
Over-ripe	46	44	47	47
Total samples	147	141	150	149

The statistical analysis in this section is intended to identify the best features to classify oil palm FFBs into certain maturity categories. The mean and standard deviations for each thermal parameter in the categories of under-ripe, ripe, and over-ripe for the first and second experiments are given in Tables 3 and 4, respectively. The two mentioned standard deviations provide information about the distribution of the relevant data. The mean values for the under-ripe, ripe, and over-ripe categories were determined to be 2.8, 1.6, and 1.2 °C, respectively, in the first experiment, while they were 3.1, 2.2, and 1.4 °C in the second experiment.

Table 3. Descriptive statistics of ΔTemp and $\Delta\text{Temp}/\text{Wt}$ (first experiment).

		N	Mean °C	Std. Deviation	Std. Error
ΔTemp	Under-ripe	44	2.88	0.485	0.070
	Ripe	53	1.83	0.248	0.030
	Over-ripe	44	1.14	0.195	0.020
	Total	141	1.94	0.76	0.060
$\Delta\text{Temp}/\text{Wt}$	Under-ripe	44	0.26	0.091	0.013
	Ripe	53	0.18	0.069	0.009
	Over-ripe	44	0.12	0.066	0.008
	Total	141	0.19	0.092	0.007

Table 4. Descriptive statistics of ΔTemp and $\Delta\text{Temp}/\text{Wt}$ (second experiment).

		N	Mean (°C)	Std. Deviation	Std. Error
ΔTemp	Under-ripe	49	3.19	0.810	0.110
	Ripe	52	2.07	0.203	0.020
	Over-ripe	47	1.20	0.338	0.040
	Total	148	2.10	0.937	0.070
$\Delta\text{Temp}/\text{Wt}$	Under-ripe	49	0.26	0.098	0.014
	Ripe	52	0.18	0.070	0.009
	Over-ripe	47	0.11	0.066	0.008
	Total	148	0.19	0.096	0.007

The results of normality tests, namely the Kolmogorov–Smirnov and Shapiro–Wilk tests, are shown in Table 5. These tests are applied to assess the normal distribution of data. According to the results, the p values for ΔTemp are greater than the selected alpha level of significance (0.05). It shows that the ΔTemp is distributed in a normal way while the $\Delta\text{Temp}/\text{Wt}$ with $p < 0.05$ needs to be altered.

Table 5. Kolmogorov–Smirnov and Shapiro–Wilk test for ΔTemp and $\Delta\text{Temp}/\text{Wt}$ data.

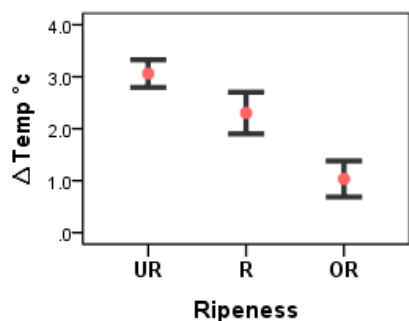
		Kolmogorov–Smirnov			Shapiro–Wilk		
		Statistic	df	Significant (<i>p</i> -Value)	Statistic	df	Significant (<i>p</i> -Value)
First experiment	ΔTemp	0.104	141	0.100	0.952	141	0.300
	$\Delta\text{Temp}/\text{Wt}$	0.101	141	0.001	0.932	141	0.007
Second experiment	ΔTemp	0.141	148	0.110	0.931	148	0.128
	$\Delta\text{Temp}/\text{Wt}$	0.110	148	0.070	0.925	148	0.080

The results of the test of homogeneity demonstrated that the assumption of homogeneity of variance (that is, Levene’s statistic test) was challenged and violated in the present study (Table 6). According to the given hypothesis, the variance of error in a dependent variable is considered to be equal across the involved groups. On the other hand, a *p*-value of ΔTemp less than 0.05 highlights the fact that there is a difference between the ranges in variance.

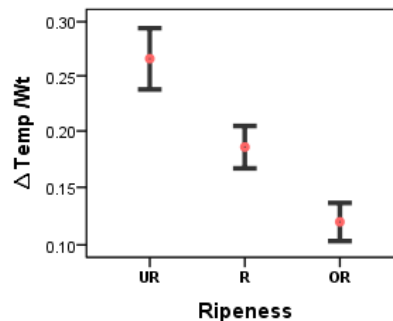
Table 6. Homogeneity test (Levene’s test) results for ΔTemp and $\Delta\text{Temp}/\text{Wt}$ data.

		Levene’s Statistics	df1	df2	Significant (<i>p</i> -Value)
First experiment	ΔTemp	20.913	2	138	0.000
	$\Delta\text{Temp}/\text{Wt}$	6.965	2	138	0.070
Second experiment	ΔTemp	44.386	2	145	0.000
	$\Delta\text{Temp}/\text{Wt}$	8.682	2	145	0.060

The error bars of ΔTemp and $\Delta\text{Temp}/\text{Wt}$ data are given in Figure 9, which contains the mean values for the parameters ΔTemp and $\Delta\text{Temp}/\text{Wt}$ for each category of the oil palm FFB. According to Figure 8, the range represents the confidence interval that relates to these means. It can be inferred that ΔTemp does not overlap with the categories of under-ripe, ripe, and over-ripe. In other words, there is no overlapping relationship among distinct groups. In the case of $\Delta\text{Temp}/\text{Wt}$, we can infer that $\Delta\text{Temp}/\text{Wt}$ does overlap between the categories of under-ripe, ripe, and over-ripe. However, if the means of $\Delta\text{Temp}/\text{Wt}$ are not significantly different, an overlapping relationship among the categories of FFBs will be expected. The ANOVA test was conducted to confirm this assumption.



(a)



(b)

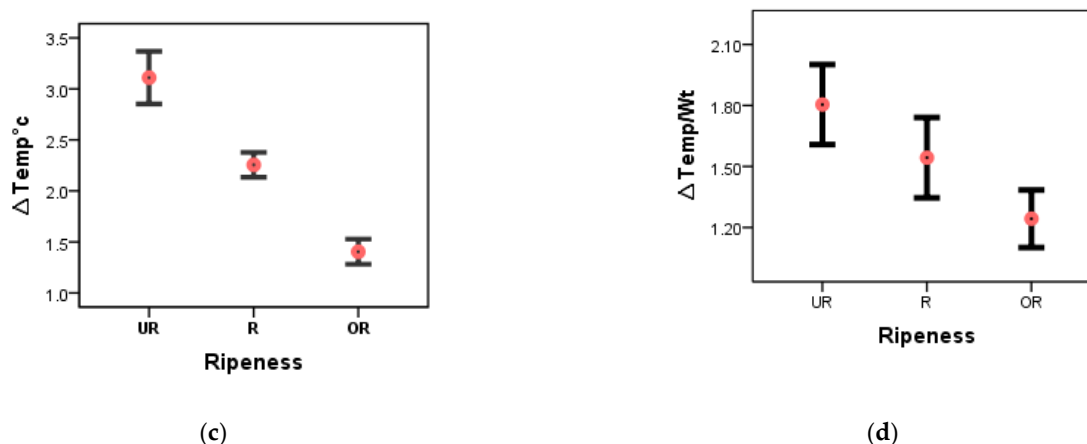


Figure 9. Error bars of $\Delta Temp$ and $\Delta Temp/Wt$: (a,b) first experiment, (c,d) second experiment.

4.2. Comparing the Mean of Temperature Variables Based on Maturity Categories

In this section, the correlation between the mean temperature measured by the thermal imaging method and the oil palm FFB maturity categories is addressed. The ANOVA test was done in order to evaluate the relationships between oil palm FFBs in the given categories and the temperature, which was determined by thermal image processing. The independent variable in the present study is the oil palm FFB maturity, which includes three groups. On the other hand, the dependent variables consisted of $\Delta Temp$ and $\Delta Temp/Wt$ derived from the thermal image processing. Both the ANOVA test and post-hoc analyses help to determine whether the data among the maturity categories are different in a significant way or not.

A summary of the results of the ANOVA test is presented in Table 7. According to these results, the significant value is believed to be less than 0.05. Therefore, these ANOVA test results have a great impact on the oil palm FFB maturity categories. The results show that the significant value for $\Delta Temp$ is below 0.05, and therefore, there is a statistically significant difference in the means of all three maturity categories. However, the means of $\Delta Temp/Wt$ were not different among under-ripe, ripe, and over-ripe (significant value > 0.05). Based on the ANOVA test results, we concluded that the variable $\Delta Temp$ is a better option than $\Delta Temp/Wt$, and we decided to use $\Delta Temp$ and drop $\Delta Temp/Wt$ from further analysis. The ANOVA test shows the difference among groups and determines which specific groups differed; we need to apply post-hoc comparison. The results of the post-hoc comparison demonstrated that there was a significant difference between all maturity categories in terms of ‘means’. Lastly, the results of the LSD and Tukey HSD tests highlighted a similar difference in the means of all maturity categories.

Table 7. ANOVA results comparing the mean of $\Delta Temp$ and $\Delta Temp/Wt$.

			Mean Square	F	Signification
First experiment	$\Delta Temp$	Between groups	33.861	312.174	0.000
		Within groups	0.108		
	$\Delta Temp/Wt$	Between groups	0.222	40.540	0.090
		Within groups	0.005		
Second experiment	$\Delta Temp$	Between groups	45.178	168.350	0.000
		Within groups	0.268		
	$\Delta Temp/Wt$	Between groups	0.257	43.161	0.600
		Within groups	0.006		

4.3. Δ Temp Behavior during the FFB Maturity Process

The results obtained from the statistical analysis indicated that the content of Δ Temp in oil palm FFB tends to decrease with the maturity stage. The bunches at the under-ripe stage turned out to have the highest index value for the parameters of Δ Temp (Figure 10). Following that, the index value underwent a decrease as the bunches ripened and decreased further when the bunches became over-ripe.

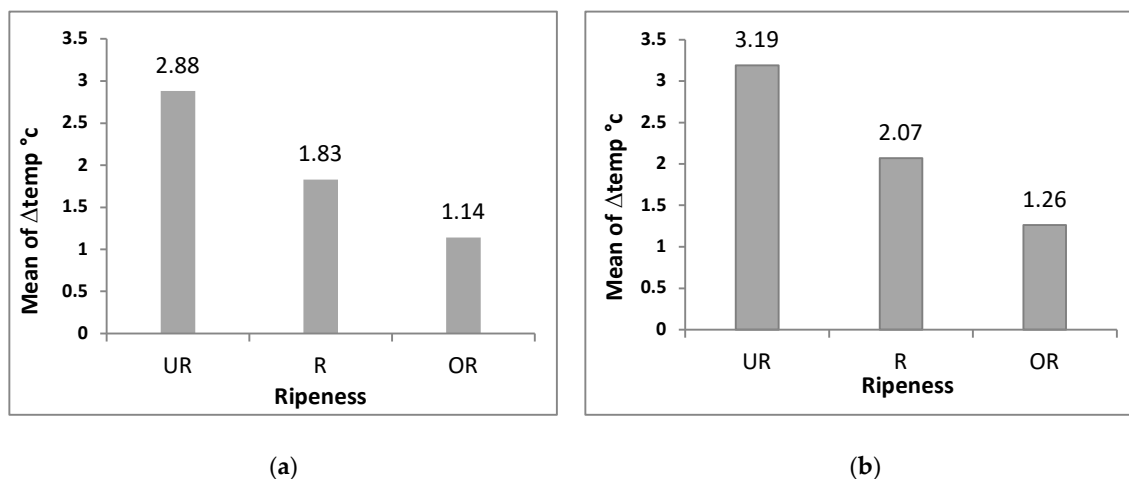


Figure 10. Mean values of Δ Temp based on different ripeness categories: (a) first experiment, (b) second experiment.

In the first experiment, the average index value of the Δ Temp was 2.8 °C at the under-ripe stage. In the ripe and over-ripe stages, the given temperature dropped to 1.8 °C and 1.2 °C, respectively. A similar pattern was exhibited in the second experiment. The average index value of the Δ Temp was 3.1 °C at the under-ripe stage, decreasing to 2.0 °C at the ripe stage and a further decrease to 1.2 °C in the over-ripe stage. According to these values, the nature of the living material and the decrease in temperature with regard to time can be specified [35]. The authors of [7] argued that biological activities at phenolic maturity are much more vigorous at the immature stage, and they may decrease or end at the over-ripe stage. As a result, the behavior of temperature in FFBs is in close affinity with the activities of the oil palm bunches in the course of the ripening stages.

The experiments demonstrated that the difference in temperature among the three categories of maturity ranged from 0.5 to 1 °C. Therefore, according to the difference in the temperatures among the under-ripe, ripe, and over-ripe fruit bunches, it can be contended that thermal imaging technology is both useful and reassuring to estimate the level of maturity.

4.4. Mean Comparison of the First and Second Experiment

The *t*-test was done to evaluate the relationship between the means of the first and second experiments on the Δ Temp factor. It revealed that the *p*-value was greater than the five percent levels (*p*-value > 0.05) of significance (5.78 > 0.05) and the *t*-test was smaller than the critical two-tail (−5.73 < 1.97). There was no significant difference between the means of the first and second experiments with regard to Δ Temp factor. It means that both experiments yielded repeatable results.

4.5. Classification Analysis of FFB Maturity

Considering Δ Temp as the major parameter of the thermal image and according to the results of the error bar and ANOVA test, Δ Temp was used for classification analysis.

The classification results of LDA, MDA, KNN, and KNN methods are discussed as follows.

4.5.1. Discriminant Analysis Classifier

The classification results of the LDA method are given in Table 8. The results of the LDA method demonstrated that the accuracy of the test stage for under-ripe, ripe, and over-ripe categories was 90.9%, 92.3%, and 76.5%, respectively. The ripe category can be recognized better in comparison with other categories. Very interestingly, no misclassification was noted in the training stage. In addition, misclassification in the test stage for the three categories was lower than that of the other classifiers. Considering the under-ripe category in the test stage, only one sample was misclassified as a false negative. The observed samples were considered to be under-ripe while the predicted ones were ripe. Similarly, in the ripe category, a total of one sample was classified as a false-negative. The number of false-negatives for the over-ripe category in the test stage was higher than for other categories.

Table 8. The classification accuracy of the training stage and test stage for LDA.

Maturity	Classification Accuracy (%)			
	Under-Ripe	Ripe	Over-Ripe	Overall
Training stage	76.3	100.0	80.0	86.0
Test stage	90.9	92.3	76.5	85.4

The classification results of the MDA method are given in Table 9. The results of the MDA method demonstrate that the accuracy of the test stage for the under-ripe, ripe, and over-ripe categories was 81.8%, 89.5%, and 71.4%, respectively. The results obtained from MDA (Table 9) indicate that the over-ripe has more misclassifications in comparison with the other categories. The ripe and under-ripe categories of MDA displayed more misclassifications in comparison with the LDA method.

Table 9. The classification accuracy of the training stage and test stage for MDA.

Maturity	Classification Accuracy (%)			
	Under-Ripe	Ripe	Over-Ripe	Overall
Training stage	84.2	91.4	84.4	86.7
Test stage	81.8	89.5	71.4	81.8

4.5.2. KNN Classifier

The classification accuracy of the KNN classifier in the training and test stages and the overall view is given in Table 10. The results of the KNN method showed that the accuracy of the test stage for under-ripe, ripe, and over-ripe categories was 87.5%, 61.5%, and 80%, respectively. The overall classification accuracy of KNN was calculated to be 82% and 72.4% for the training and testing stages, respectively. The KNN method produced the highest correct classification in the under-ripe category, at 100% for the training stage. Despite that fact, KNN was considered to be favorable for the under-ripe and ripe categories while it was not satisfactory for the over-ripe category.

Table 10. The classification accuracy of the training stage and test stage for KNN.

Maturity	Classification Accuracy (%)			
	Under-Ripe	Ripe	Over-Ripe	Overall
Training stage	100.0	79.4	60.9	82.0
Test stage	87.5	61.5	80.0	74.2

4.5.3. ANN Classifier

Table 11 contains the accuracy of correct classification for the ANN classifier in the training and test stages. The results of the ANN method demonstrated that the accuracy of the test stage for under-ripe, ripe, and over-ripe categories was 100%, 90%, and 86%, respectively. The accuracy for the overall correct classification of the ANN was 99.1% and 92.5% for the training and test stages, respectively. The lift and gain diagrams (Figure 11) are graphical aids for evaluating the classification performance model. According to the curve position and baseline, it can be concluded that the ANN model shows a higher overall gain and indicates perfect performance for all three categories. Based on the results of the ANN classifier, the under-ripe category can be differentiated from the other categories in a more effective way. Both in the training and the testing stages, there appeared to be no misclassification. The ripe and over-ripe categories encountered a number of errors. In the ripe and over-ripe categories, only one sample was misclassified as false negative in the testing stage. The observed samples were ripe while the predicted ones were considered to be under-ripe. In the over-ripe category, one sample was falsely predicted to be in the ripe category.

Table 11. The classification accuracy of the training stage and test stage for ANN.

Maturity	Classification Accuracy (%)			
	Under-Ripe	Ripe	Over-Ripe	Overall
Training stage	100.0	97.4	100.0	99.1
Test stage	100.0	90.9	86.7	92.5

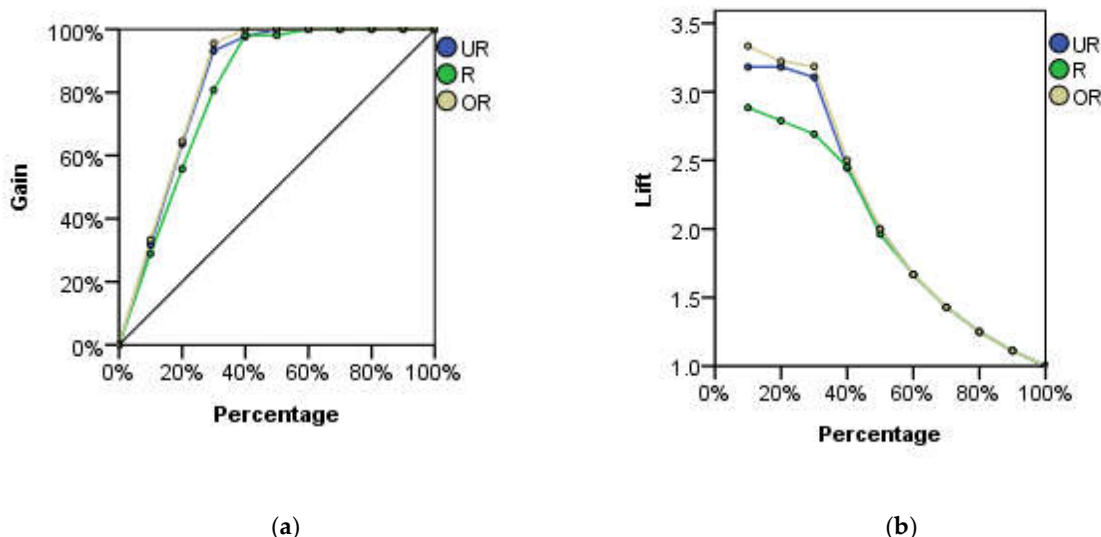


Figure 11. ANN performance measurement (a) cumulative gains chart (b) lift chart.

4.5.4. Classification Results Comparison

Table 12 shows the accuracy of four different methods of classification. It should be noted that the Δ Temp predictor provided by thermal image processing played a key role in this part. The average accuracy of overall classification in LDA, MDA, ANN, and KNN was 86%, 86.7%, 99.1%, and 82.2% for training stages and 85.4%, 81.8%, 92.5%, and 74.3% for test stages, respectively. The results showed that the ANN method gave the highest overall correct classification accuracy (99.1% and 92.5%) compared to other classification methods.

Table 12. Accuracy comparison of classification methods in the training stage and test stage.

Overall Percent (%) Correct Classification Based on Different Methods				
	LDA	MDA	ANN	KNN
Training stage	86.0	86.7	99.1	82.0
Test stage	85.4	81.8	92.5	74.2

It can be concluded that ΔTemp with regard to the ANN method is more accurate compared to other previous models. In a previous work on oil palm maturity classification, the HSI color model based on the ANN algorithm [36] and the color model with SVM [37] both produced high correct classification accuracy with 94% and 96.95%, respectively. The accuracy of the RGB color model with the SVM method is higher because it is a bi-class classifier method and it classified oil palm FFBs into under-ripe and ripe categories. The thermal imaging with the ANN classifier can achieve 100% accuracy both in the training and the testing stages of the under-ripe category, and in the training stage of the over-ripe category, there appeared to be no misclassification. Furthermore, it produces high classification accuracy (97.4%) in the ripe category. It improved oil palm FFB maturity classification accuracy due to the high correlation between ΔTemp obtained by thermal imaging and oil palm FFB maturity.

The main drawback of RGB color methods is that they are affected by changing light intensities. The important information about the oil palm bunches became obscured in the images due to illumination and noise. HSV is less affected by illumination changes compared to RGB. However, thermal imaging with the ANN classifier method is not affected by lighting as compared with challenges faced in imaging required for color models.

This research makes a new contribution in the classification of maturity of oil palm FFBs through the use of the ΔTemp parameter which is computed based on thermal imaging data, together with ANN. It is a fast and simple method with high classification accuracy for grading FFBs of oil palm.

5. Conclusions

The maturity of oil palm fruits will affect the quality of palm oil. Classifying the oil palm FFBs into maturity categories is an important factor because it will make milling operations more efficient. The purpose of this research work was to address the role of a thermal imaging technique to quantify the oil palm FFB maturity.

The thermal images of FFB were processed to calculate the overall temperature of the oil palm bunches. The difference between oil palm FFB temperature and the ambient temperature, namely, ΔTemp , was considered to be the main contributing parameter. The ΔTemp of under-ripe, ripe, and over-ripe were significantly different among the oil palm fruit bunch maturity categories. It can be concluded that ΔTemp is a reliable index to classify the FFBs of oil palm. The key findings rely on the fact that the mature fruits have a higher temperature than the immature ones. Therefore, the bunches at the under-ripe stage turned out to have the highest index value for the parameters of ΔTemp , and the over-ripe category has the lowest value of ΔTemp . Therefore, based on the difference in temperature of under-ripe, ripe, and over-ripe fruits, the thermal imaging technology is very promising in estimating the maturity level of the FFB.

Four classification methods, namely LDA, MDA, KNN, and ANN, were used to categorize oil palm FFBs into the following three categories: under-ripe, ripe, and over-ripe. The overall classification accuracy was 85.4%, 81.8%, 92.5%, and 74.3% for LDA, MDA, ANN, and KNN, respectively. The highest degrees of overall accuracy of 99.1% (Training) and 92.5% (Testing) were obtained through the ANN method. Hence, the ANN was evaluated as the best prediction method to categorize oil palm FFBs into the three maturity groups. The purpose of this research work was to address the role of thermal imaging techniques in quantifying the oil palm FFB maturity, OC, and oil quality

parameters. The novelty of this method is that thermal imaging was used instead of RGB images and temperature was considered as the main parameter to classify oil palm FFB maturity. RGB color methods are suitable to be used in a constant lighting environment since they are affected by illumination and noise. However, thermal imaging is less affected by the lighting environment compared to color models. It produced a non-contact, non-destructive, fast, and simple in-field method with acceptable classification accuracy (more than 90%) for grading FFBs of oil palm.

Author Contributions: Conceptualization, S.Z. and A.R.B.M.S., method, S.Z. and A.R.B.M.S., software, S.Z., validation, S.Z. and A.R.B.M.S., writing—original draft preparation, S.Z. and A.R.B.M.S., Writing—review and editing, A.R.B.M.S., H.Z.J., R.E., and I.B.A., supervision, A.R.B.M.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was conducted under the Long-Term Research Grant Scheme (LRGS) from the Ministry of Higher Education of Malaysia (MOHE) bearing an institutional vote number of 5526100.

Data Availability Statement: The data are available on request from the corresponding author. The data are not publicly available due to privacy restrictions.

Acknowledgments: This research was conducted under the Long-Term Research Grant Scheme (LRGS) from the Ministry of Higher Education of Malaysia (MOHE) bearing an institutional vote number of 5526100. The authors gratefully appreciate this trust and funding from MOHE. We record our special appreciation to the Universiti Putra Malaysia for providing the research and support infrastructure, to the United Plantation Research Department (UPRD), for their valuable help, support, cooperation, and contributions. We are also thankful to all those who have contributed to this project by providing valuable inputs.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. MPOB. Overview of The Malaysian Oil Palm Industry. From Economic & Industry Development Division. 2021. Available online: <http://www.bepi.mpob.gov.my/index.php/en/> (accessed on 20 October 2022).
2. Kassim, M.S.; Ismail, W.I.; Ramli, A.R.; Bejo, S.K. Oil palm fresh fruit bunches (FFB) growth determination system to support harvesting operation. *J. Food Agric. Environ.* **2012**, *10*, 620–625.
3. Vijaya., S.; Ma, A.N.; Choo, Y.M. A Gate to Gate Assessment of Environmental Performance for Production of Crude Palm Kernel Oil Using Life Cycle Assessment Approach. *Am. J. Environ. Sci.* **2009**, *5*, 267–272.
4. Junkwon, P.; Takigawa, T.; Okamoto, H.; Hasegawa, H.; Koike, M.; Sakai, K.; Siruntawinetti, J.; Chaeychomsri, W.; Vanavichit, A.; Tittinuchanon, P.; Bahalayodhin, B. Hyperspectral imaging for nondestructive determination of internal qualities for oil palm (*Elaeis guineensis* Jacq. Var. tenera). *Agric. Inf. Res.* **2009**, *18*, 130–141.
5. Saeed, O. M.B.; Shariff, A.R.M.; Shafri, H.Z.M.; Mahmud, A.R.; Alfatni, M.S.M. Hyperspectral Technique System for Fruit Quality Determination. Presented at the Map Asia 2010 & ISG 2010, Kuala Lumpur, Malaysia, 26–28 July 2010.
6. Hazir, M.H.M. Ripeness Detection of Oil Palm Fresh Fruit Bunches Using Fluorescence Sensor. Master’s Thesis, University Putra Malaysia, Selangor, Malaysia, 2011.
7. Hazir, M.H.M.; Shariff, A.R.M.; Amiruddin, M.D. Determination of oil palm fresh fruit bunch ripeness Based on flavonoids and anthocyanin content. *Ind. Crops Prod.* **2012**, *36*, 466–475.
8. Alfatni, M.S.; Shariff, A.R.; Abdullah, M.Z.; Marhaban, M.H.; Shafie, S.B.; Bamiruddin, M.D.; Saeed, O.M. Oil palm fresh fruit bunch ripeness classification based on rule-based expert system of ROI image processing technique results. *IOP Conf. Ser. Earth Environ. Sci.* **2014**, *20*, 012018.
9. Makky, M.; Sony, P.; Salokhe, V. Automatic Non-destructive Quality Inspection System for Oil Palm Fruits. *Int. Agrophysics J.* **2014**, *28*, 319–329. <https://doi.org/10.2478/intag-2014-0022>.
10. Harun, N.; Mirson, N.; Roslina, S.; Ishak, A.; Hiroyuki, W.; Kunihiisa, T. Dual Resonant Frequencies Effects on an Induction-Based Oil Palm Fruit Sensor. *Sensors* **2014**, *14*, 21923–21940. <https://doi.org/10.3390/s141121923>.
11. Mirson, N.; Harun, N.; Yeoh, L.; Roslin, S.; Ishak, A.; Hiroyuki, W.; Kunihiisa, T. Improvement in Sensitivity of an Inductive Oil Palm Fruit Sensor. *Sensors* **2014**, *14*, 2431–2448. <https://doi.org/10.3390/s140202431>.
12. Silalahi, D.; Consorcia, E.R.; Lansigan, F.P.; Panopio, R.G.; Bantayan, N.; Caliman, J.P.; Yuan, Y.Y.; Sudarno, S.; Davrieux, F. Near-infrared spectroscopy: A rapid and non-destructive technique to assess the ripeness of oil palm (*Elaeis guineensis* Jacq.) fresh fruit. *J. Near Infrared Spectrosc.* **2016**, *24*, 179–190. <https://doi.org/10.1255/jnirs.1205>.
13. Shiddiq, M.; Sala Mbue, R.Z.; Yasmin, N.; Lismayeni, F.; Fitridhani, S.; Adzani, H. Laser based imaging method to discriminate Riau Province pure honeys. *J. Phys. Conf. Ser.* **2018**, *1120*, 012053. <https://doi.org/10.1088/1742-6596/1120/1/012053>.

14. Zuhaira., M.Z.; Fazida., H.; Thinal, R.; Aqilah, B.H. A Rapid and Non-Destructive Technique in Determining the Ripeness of Oil Palm Fresh Fruit Bunch (FFB). *J. Kejuruter.* **2018**, *30*, 93–101. <https://doi.org/10.17576/jkukm-2018-30-12>.
15. Alfatni, M.S.M.; Shariff, A.R.M.; Shafri, H.Z.M.; Saaed, O.M.B.; Eshanta, O.M. Oil palm fruit bunch grading system using red, green and blue digital numbers. *J. Appl. Sci.* **2008**, *8*, 1444–1452.
16. Ishak, W.I.W.; Zohadie, M.B.; Hamid, A.M.A. Optical Properties for Mechanical Harvesting of Oil Palm FFB. *J. Oil Palm Res.* **2000**, *12*, 38–45.
17. Saeed, O.M.B.; Shariff, A.R.M.; Shafri, H.Z.M.; Shafri, H.Z.M.; Mahmud, A.R.; Alfatni, M.S.M. Classification of oil palm fresh fruit bunches based on their maturity using portable four-band sensor system. *Comput. Electron. Agric.* **2011**, *82*, 55–60.
18. Hong, T.S.; Hashim, F.H.; Raj, T.; Huddin, A.B. Classification of Oil Palm fruit Ripeness Using Artificial Neural Network. In Proceedings of the IEEE International Conference On Automatic Control & Intelligent System (I2CACIS), Shah Alam, Malaysia, 26 June 2021; pp. 358–363. <https://doi.org/10.1109/I2CACIS52118.2021.9495857>.
19. Robi, S.N.A.B.M.; Izhar, M.A.B.M.; Ahmad, N.B. Image Detection and Classification of Oil Palm Fruit Bunches. In Proceedings of the 4th International Conference on Smart Sensors and Application (ICSSA), Kuala Lumpur, Malaysia, 26–28 July 2022; pp. 108–113. <https://doi.org/10.1109/ICSSA54161.2022.9870945>.
20. Tzuan, G.T.H.; Hashim, F.H.; Raj, Baseri Huddin, A.; Sajab, M.S. Oil Palm Fruits Ripe-ness Classification Based on the Characteristics of Protein, Lipid, Carotene, and Guanine/Cytosine from the Raman Spectra. *Plants* **2022**, *11*, 1936. <https://doi.org/10.3390/plants11151936>.
21. Hellebrand, H.J.; Linke, M.; Beuche, H.; Herold, B.; Geyer, M. Horticultural products evaluated by thermography. In *AgEng*; Paper No. 00-PH-003; University of Warwick: Coventry, UK, 2000.
22. Vadivambal, R.; Jayas, D.S. Applications of thermal imaging in agriculture and food industry-a review. *Food Bioprocess Technol.* **2011**, *4*, 186–199.
23. Danno, A.; Miyazato, M.; Ishiguro, E. Quality evaluation of agricultural products by infrared imaging method. III. Maturity evaluation of fruits and vegetables. *Mem. Fac. Agr. Kagoshima Univ.* **1980**, *16*, 157–164.
24. Hahn, F.; Hernández, G.; Echeverría, E.; Romanchick, E. Escherichia coli detection using thermal images. *Can. Biosyst. Eng.* **2006**, *48*, 4.7–4.13.
25. Ishimwe, R.; Abutaleb, K.; Ahmed, F. Applications of thermal imaging in agriculture—A review. *Adv. Remote Sens.* **2014**, *3*, 128.
26. Baranowski, P.; Mazurek, W.; Walczak, B.W.; Sławiński, C. Detection of early apple bruises using pulsed-phase thermography. *Postharvest Biol. Technol.* **2009**, *53*, 91–100.
27. Gan-Mor, S.; Regev, R.; Levi, A.; Eshel, D. Adapted thermal imaging for development of postharvest precision steam – disinfection technology for carrots. *Postharvest Biol. Technol.* **2011**, *59*, 265–271.
28. Baranowski, P.; Mazurek, W.; Wozniak, J.; Majewska, U. Detection of Early Bruises in Apples Using Hyperspectral Data and Thermal Imaging. *J. Food Eng.* **2012**, *110*, 345–355. <https://doi.org/10.1016/j.jfoodeng.2011.12.038>.
29. Sumridetchkajorn, S.; Intaravanne, Y. Two-Dimensional Fruit Ripeness Estimation using Thermal Imaging. In Proceedings of the International Conference on Photonics Solutions, Pattaya City, Thailand, 7 June 2013; Volume 8883, CCC Code: 0277-786X/13/\$18. <https://doi.org/10.1117/12.2019654>.
30. Bejo, K.S.; Azman, N.; Jamil, N. Paddy Grading Using Thermal Imaging Technology. *Int. Food Res. J.* **2016**, *23*, S245–S248.
31. Mohd Ali, M.; Hashim, N.; Abd Aziz, S.; Lasekan, O. Characteriation of Pineapple cultivares under Different Storage Condition Using Infrared Thermal Imaging Coupled with Machine Learning Algorithms. *Agriculture* **2022**, *12*, 1013. <https://doi.org/10.330/agriculture12071013>.
32. Malaysian Palm Oil Board. *Manual Penggredan Buah Kelapa Sawit*, 3rd ed.; Malasian Palm Oil Board (MPOB): Kajang, Malaysia, 2006.
33. Zolfagharnassab, S.; Shariff, A.R.M.; Ehsani, R. Emissivity Determination of Oil Palm Fresh Fruit Ripeness Using A Thermal Imaging Technique. *Acta Hort.* **2017**, *1152*, 189–194. <https://doi.org/10.17660/ActaHortic.2017.1152.26>.
34. Budzan, S.; Wyżgolik, R. Remarks on Noise Removal in Infrared Images. *Meas. Autom. Monit. J.* **2015**, *61*, 6.
35. Dey, P.M.; Brownleader, M.D.; Harborne, J.B. Theplant, the cell and its molecu-lar components. In *PlantBiochemistry*; Dey, P.M., Harborne, J.B., Eds.; Academic Press: London, UK, 1997; pp. 1–47.
36. Fadilah, N.O.; Mohamad-Saleh, J.U. Color feature extraction of oil palm fresh fruit bunch image for ripeness classification. In Proceedings of the 13th International Conference on Applied Computer and Applied Computational Science (ACACOS'14), Kuala Lumpur, Malaysia, 23–25 April 2014; pp. 51–55.
37. Sabri, N.; Ibrahim, Z.; Syahlan, S.; Jamil, N.; Mangshor, N.N.A. Palm Oil Fresh Fruit Bunch Ripeness Grading Identification Using Color Feature. *J. Fundam. Appl. Sci.* **2017**, *9*, 563–579.